20/8/26

Exp: 9 Implementation of Subnetting in Cisco Packet Tracer.

Aim:

To implement classless IP subnetting and configure network devices using Cisco Packet Tracer.

Procedure:

- Create a new network topology in Cisco packet tracer.
- Add routers, switches, and PCs then connect them using suitable cables.
- Divide router interface with IP address and cable 192.168.1.0/24 into smaller subnets using /27 (255.255.255.224).
- Configure router interface with IP address and enable them using no shut down.
5. Assign static IP addresses, subnet masks, and default gateways to all PCs.
- Network - 192.168.1.0/27
  Router IP - 192.168.1.1
  PC IP Address - .11 - .15
  Subnet mask : 255.255.255.224
- Router configuration commands.
  enable, configure terminal, interface FastEthernet

ip address 192.168.1.1 255.255.224, no shutdown.

- configure the remaining interfaces with their respective IP addresses.

Testing:

- Open the Command prompt on each PC.
- Use the ping command to test connectivity.
- Verify that the devices can communicate

Result:

Thus, classless IP subnetting was implemented successfully using Cisco Packet Tracer, and network connectivity was tested using the ping command.